# Home Credit Default Risk

Payment difficulty across 307,511 granted loans · SQL analysis of application, bureau and previous-application data

Overview | Applicants | Affordability | Credit history

## 8.07% of loans had payment difficulties. Overdue credit, age and job tenure separate risk best; debt-burden ratios barely do.

Above the 8.07% portfolio rate | At or below it | Portfolio rate | All bars share one 0 to 18% scale

| LOANS ANALYSED | PAYMENT DIFFICULTY CASES | PAYMENT DIFFICULTY RATE |
|---|---|---|
| 307,511 | 24,825 | 8.07% |
| Granted loans with a known outcome | Loans flagged TARGET = 1 | About 1 in 12 loans |

Every row is a loan that was already granted. Payment difficulty is the dataset's TARGET = 1 flag: the client had difficulty repaying. Rates are shares of loans, not money lost.

### Key findings

**Overdue credit is the clearest warning.**

Applicants with an overdue bureau credit had a 15.90% rate, about 2.1× the 7.62% of applicants with a clean record. They are only 1.1% of loans.

**Young and short-tenure applicants carry more risk.**

Ages 20 to 30 run at 11.47% against 4.92% at 60 to 70. Applicants employed 0.4 to 2.3 years reach 11.28%.

**Affordability ratios say little on their own.**

Credit ÷ income bands span 6.43% to 9.11%, and annuity ÷ income bands 7.27% to 8.77%.

**A strong external score cuts the rate by about 60%.**

Applicants with an external score above 0.66 ran at 3.8% against 9.6% for everyone else, including those with a high credit ÷ income ratio.

### Which factors move the rate most?

Table

Gap between the highest and lowest group, in percentage points. Each bar spans lowest to highest rate.

Overdue bureau credit 7.62% to 15.90% **8.3 pts**

Age 4.92% to 11.47% **6.6 pts**

| | | |
|---|---|---|
| Years employed | 5.37% to 11.28% | 5.9 pts |
| External score (above or below 0.66) | 3.83% to 9.56% | 5.7 pts |
| Previous refusals | 6.98% to 12.61% | 5.6 pts |
| Education | 5.36% to 10.93% | 5.6 pts |
| Loan size | 4.43% to 9.46% | 5.0 pts |
| Contract type | 5.48% to 8.35% | 2.9 pts |
| Prior bureau loans | 7.40% to 10.12% | 2.7 pts |
| Credit ÷ income | 6.43% to 9.11% | 2.7 pts |
| First-time or repeat | 5.96% to 8.19% | 2.2 pts |
| Income level | 6.82% to 8.69% | 1.9 pts |
| Annuity ÷ income | 7.27% to 8.77% | 1.5 pts |
| Owns realty | 7.96% to 8.32% | 0.4 pts |

Groups under 1,000 loans are ignored. Occupation is left out because only the ten highest-rate occupations were queried.

## Rate by loan size

Table

Credit amount of the current loan

| | | |
|---|---|---|
| Under 250K | 59,198 loans | 7.13% |
| 250K–500K | 90,135 loans | 9.30% |
| 500K–750K | 65,094 loans | 9.46% |
| 750K–1M | 43,099 loans | 7.26% |
| 1M–1.5M | 39,229 loans | 6.26% |

1.5M and above 10,756 loans 4.43%

Risk peaks at 250K to 750K (9.3% to 9.5%) and falls to 4.43% above 1.5M.

## Rate by contract type

Table

Cash loans 278,232 loans 8.35%

Revolving loans 29,279 loans 5.48%

Revolving loans run 2.9 points below cash loans.

## Borrowers with and without difficulty

Average income, credit amount and annuity, in the dataset's currency units

| Average | No difficulty | Difficulty | Gap |
|---|---|---|---|
| Income | 169,078 | 165,612 | −2.0% |
| Credit amount | 602,648 | 557,779 | −7.4% |
| Annuity | 27,164 | 26,482 | −2.5% |

Borrowers with difficulty earned and borrowed slightly less, so size and income alone do not separate them.